RKDF University, Bhopal
Faculty Profile

Basic Information	
Name	Dr. Balajee Sharma
Date of Birth	03/04/1992
Designation	Associate Professor
Department	Electronics & Communication Engineering
Experience	10 Year 10 Month
Email ID	sharmabalajee@gmail.com
Contact No	9039437620

Educational Qualifications

Description	Year	%	Institute/University
(UG)-	2013	66.14	RGPV Bhopal
(PG)-	2015	85.00	RKDF University Bhopal
M. Phil.			
(Ph. D.)-	2022		RKDF University Bhopal
Post Doctorate			
NET Qualified/GATE			

Experience Detail

Experience (Teaching/Research)	Designation	Duration		Name of Institute/University
		From	To	
Teaching/Research	Associate Professor	September 2013	May 2025	RKDF University

Publications

No. of Papers Published (Attach Annexure of papers published in peer-reviewed or refereed journals)	15
No. of Books Published	-
Books Chapters Published (Provide print of 1st Page of Book	-
No. of Patents Published/Grant	2

Ph. D/M. Phil Project supervised			
Research Program	Award	Under Supervision	Name of University
Ph. D (Provide detail i.e. name, title etc)		01	RKDF University
M. Phil	-		
PG Thesis/Dissertation	-		

Area of Expertise (100 words)

Award and Achievement	
Name of Award	Description (With certified of Copy award)
National	-
International	-

Conference/Seminar/Workshops/FDP	
Description	No.
Conference/Seminar paper presentation	6
Conference/Seminar attended/ organized	6
Work shop attended/ organized	-
FDP Attended/ organized	-

Research Project		
Name of Project	Funding Agencies	Amount
-	-	-

- Any other Achievement

T. Jaiajee Sharma

Signature

(16)

List of Regular faculty with Designation, who are eligi[illegible] as per UGC Ph.D. regulation 2022 or relevant Reg[illegible]rity Council norms for recognition as Ph.D./ Research Supervisor in the format given below:

(Please attach prescribed format with relevant documents)

Sr. no.	Name of the Regular faculty	Designation and Pay Scale	**Date of appointment in University	Date of birth	Qualification	Specialization	**Ph.D award Year	**Total Experience as Regular faculty in years		**No. of research publications in Peer reviewed or refereed Journals	No. of Ph.D. candidates awarded under supervision	Vacant Seats for Ph.D. Research Teaching supervision
								Research	Teaching			
1.	Dr. Balajee Sharma	Associate Professor	26 Sep 2013	03.04 .1992	P.hD, M.Tech, B.E	Electronics & Communication Engineering	2022	9 Year	11Year 7 Months	15	01	05

Note Please enclose all relevant document**

Number of Seminar/Workshop/Conference attended.

During last three years:

a. International Level- ____05____
b. National Level-________
c. State Level- ________
d. University Level- ________

Number of Research Papers & Books published

a. International Level:____15____
b. National Level:________
c. State/Regional/University Level:________

Curriculum Vitae

DR. BALAJEE SHARMA
Contact: +91-9039437620; 9425
E-mail: sharmabalajee@gmail.com; balajee982@gmail.com
Address: Flat No-603 Wing A Tirupati ML Highrise Ayodhya Bypass Road Bhopal (M.P) India 462010

Area of specialization: MIMO-OFDM ,Orthogonal Frequency Division, Spectrum Sensing, Multiplexing, Frequency Index Modulation, Channel, Estimation, Deep Neural Network, DNN, AWGN

Personal Skills:

Critical Thinking, Research Skills, Leadership, Mentorship, Time Management, Cultural Sensitivity, Innovative Problem Solver, good verbal and written communication skills, analytical thinker and adaptive learner.

Objectives:

Teaching and Education:

Deliver Quality Instruction: Provide high-quality education to undergraduate and graduate students in ECE-related courses, ensuring that students gain a solid understanding of the subject matter.
Foster Learning: Create an engaging and inclusive learning environment that promotes critical thinking, problem-solving skills, and a passion for the field.
Mentor Students: Act as an advisor and mentor to students, offering guidance on academic and career matters.

Research and Scholarship:

Conduct Research: Engage in research activities in your area of expertise within ECE, conducting experiments, publishing papers, and contributing to the advancement of knowledge in the field.
Secure Funding: Seek external research funding through grants and collaborations to support your research projects.
Supervise Graduate Research: Mentor and supervise graduate students in their research work, guiding them toward successful theses and dissertations.
Publish: Disseminate research findings through peer-reviewed journals, conferences, and other scholarly publications.

Professional Development:

Stay Current: Continuously update your knowledge and expertise in ECE by staying abreast of the latest developments, technologies, and research trends.
Networking: Build and maintain relationships with colleagues, both within and outside your institution, to foster collaboration and access to resources.

Service to the Institution and Community:

Departmental Service: Serve on departmental committees, contribute to curriculum development, and participate in other activities that support the department's goals and missions.

University Service: Engage in university-wide initiatives, committees, and academic governance.

Community Engagement: Collaborate with industry, local organizations, and the community to apply ECE knowledge and expertise to real-world problems.

Professional Recognition

Seek Promotion: Work toward achieving tenure and advancing in your academic career.
Awards and Recognition: Strive for recognition in the form of awards, honors, and accolades for outstanding teaching, research, or service contributions.

Work-Life Balance:

Achieve Balance: Strive for a healthy work-life balance to maintain your well-being and sustain a long and productive career in academia.

Educational Qualifications:

- Ph.D. in Electronics& Communication Engineering from Ram Krishna Dharmarth Foundation University Bhopal in year 2022.
  Title of the thesis: "*AI Based Performance Enhancement in MIMO OFDM*"
- Master of Technology in Digital Communication from RKDF University with first division with distinction; CGPA 8.5 in Year 2015.
- Bachelor in Engineering in Electronics & Communication Engineering from Rajiv Gandhi Technical University with first division; 66.14 % in year 2013.
- Intermediate (10+2) from BSEB Patna with first division with distinction; 61.14 %. in year 2009.
- High School (10^{th}) from CBSE with Second division with.58.43 %. in year 2007.

Research Interests:

Channel Modeling and Estimation: Investigating and modeling the wireless channel in MIMO-OFDM systems, especially in complex environments like urban or indoor settings. Developing efficient algorithms for channel estimation and tracking to improve system performance.

Preceding and Beam forming Techniques: Designing and analyzing advanced precoding and beam forming schemes for MIMO-OFDM to enhance spectral efficiency and reduce interference.

Resource Allocation and Optimization: Developing resource allocation algorithms that allocate subcarriers, power, and antennas efficiently to maximize system capacity, minimize interference, or meet quality-of-service (QoS) requirements.

Interference Mitigation and Coexistence: Investigating techniques to mitigate interference, especially in dense wireless networks, such as interference-aware preceding and interference cancellation.

Massive MIMO and Beyond: Exploring the potential of massive MIMO systems, which involve a large number of antennas at both transmitter and receiver, and their application in MIMO-OFDM systems. Beyond massive MIMO, looking at new paradigms like cell-free massive MIMO and holographic MIMO.

Synchronization and Timing: Developing synchronization algorithms that ensure proper timing and frequency synchronization in MIMO-OFDM systems, even in the presence of channel impairments.
Channel Coding and Error Correction: Researching coding techniques tailored for MIMO-OFDM, such as space-time coding, to improve error correction and reliability.
Energy Efficiency and Green Communications: Investigating energy-efficient MIMO-OFDM systems, considering power allocation strategies and the impact on battery life in wireless devices.
Millimeter-Wave (mm Wave) MIMO-OFDM: Exploring MIMO-OFDM at high-frequency bands, like mm Wave, which has unique challenges and opportunities for improving data rates.
Cross-Layer Optimization: Integrating MIMO-OFDM with other communication layers, such as the network layer, for cross-layer optimization to achieve better overall system performance.
Machine Learning and AI Applications: Leveraging machine learning and artificial intelligence techniques for optimizing MIMO-OFDM system parameters, predicting channel conditions, or developing intelligent resource allocation strategies.
Security and Privacy: Investigating security challenges and privacy concerns in MIMO-OFDM systems, and developing encryption and authentication techniques to protect data transmission.
Standardization and Prototyping: Contributing to the development of MIMO-OFDM standards or implementing real-world prototypes to validate research findings.

Software Proficiency:

- MATLAB and Simulink

Working/Teaching Experience:

- Currently working as an Associate Professor at Ram Krishna Dharmarth Foundation University Bhopal. (26 September 2013 to till date)

Administrative Responsibilities:

- Working as Admission Counseling Guidance
- Working as Assistant TPO
- Other works allotted by Management

List of Research Publications/Confrence :

1. **Balajee Sharma,** Akant Kumar Raghuwanshi and Ajay Barapatre (2025) Emerging Trends in 5G and Beyond-5G Wireless Networks. Annual International Conference on Recent advanced in Engineering Technology , Healthcare& Management (AIC-RAETHM 2025) *Jan10,11 2025.*

2. Ajay Barapatre, **Balajee Sharma** and Akant Kumar Raghuwanshi (2025) A Study on Security Requirements of IOT. Annual International Conference on Recent advanced in Engineering Technology , Healthcare& Management (AIC-RAETHM 2025) *Jan10,11 2025.*

3. Akant Kumar Raghuwanshi, Ajay Barapatre and **Balajee Sharma** (2025) Wireless Sensor Networks and Its Clustering Algorithm. Annual International Conference on Recent advanced in Engineering Technology , Healthcare& Management (AIC-RAETHM 2025) *Jan10,11 2025.*

4. Hemant Rajoriya and **Balajee Sharma** (2024) A Comprehensive Review of Deep Compressive Sensing for Efficient IoT Data Management. *Journal Trends in Electrical Engineering, 14(03).*

5. Hemant Rajoriya and **Balajee Sharma** (2024) Deep Learning-Enhanced Compressive Sensing for Wireless IoT Data Optimization and Weather Monitoring. *International Journal of Satellite Remote Sensing, 02(02).*

6. Hemant Rajoriya and **Balajee Sharma** (2024) Innovative Approaches to Reducing Data Traffic in IoT Networks Using Deep Learning and Compressive Sensing. *International Journal of Satellite Remote Sensing, 02(02).*

7. **Balajee Sharma** and Akant Kumar Raghuwanshi (2024) Advancements in AI-driven Techniques for MIMO-OFDM Systems: A Comparative Analysis. *International Journal of Innovative Research in Technology and Science, 11(02).*

8. Akant Kumar Raghuwanshi and **Balajee Sharma** (2024) Investigating Power-Aware Protocols for Wireless Sensor. *International Journal of Innovative Research in Technology and Science, 11(02).*

9. **Balajee Sharma and** Virendra Singh Chaudhary (2022) Channel Estimation and Equalization Using FIM for MIMO-OFDM on Doubly Selective Faded Noisy Channels. *ecti transactions on electrical engineering, electronics, and communications vol.20, no.1 february 2022.*

10. **Balajee Sharma and** Virendra Singh Chaudhary (2021) Channel Estimation using Frequency Index Modulation for MIMO-OFDM on Doubly Selective Faded Noisy Channel. *Information Technology & Electrical Engineering (ITEE) 10 (3).*

11. **Balajee Sharma and** Virendra Singh Chaudhary (2021) Channel Estimation using Frequency ndex Modulation for MIMO-OFDM. *Journal of the Sayajirao university of Baroda. ISSN :0025-0422 55 (1).*

12. **Balajee Sharma and** Virendra Singh Chaudhary (2019) Channel Estimation using STTC Index Modulation for MIMO-OFDM. *2nd International Conference On "Contemporary Technological Solutions Towards Fulfillment Of Social Needs" BY RKDF University September 28-29, 2019.*

13. **Balajee Sharma and** Virendra Singh Chaudhary (2018) A Study of Spectrum Sensing based Channel Estimation for MIMO–OFDM over Noisy Channel. *International Conference on Contemporary Technological Solutions towards fulfillment of Social Needs" BY RKDF University August, 2018.*

14. **Balajee Sharma** and Rashmi Pandey (2015) A Performance Enhancement in BER for OFDM System Using 16-QAM Technique and Viterbi Algorithm. *international journal of scientific progress and research (ijspr),* 14(01).

15. **Balajee Sharma** and Rashmi Pandey (2015) A Review: Modelling and Performance of OFDM System using QAM. *international journal of scientific progress and research (ijspr),* 13(03).

Patents Published / Filed:

1. Dr. Rashmi Pandey, Mr. Kuldeep Pandey, Mr. Abhinav Shukla, Dr. Akant Kumar Raghuwanshi, **Dr. Balajee Shrama**, Mr. Ajay Kumar Barapatre " *Wireless power Transfer Plans for Installable Biomedical Components* " *2024. APPLICATION NO.202321064948.*

Ph.D Research Scholar Admitted

1. Yashwant Singh 2023 (Course work completed)

Declaration

I hereby declare that all the statements made in the curriculum vitae are true to the best of my knowledge and belief.

DATE:

PLACE: Bhopal

Balajee Sharma.

(Balajee Shrama)

Google Scholar Citations:

https://scholar.google.com/citations?user=CNLLp_OAAAAJ&hl=en